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Problem Statement

Question : The area between $x = y^2$ and $x = 4$ is divided into two equal parts by the line $x = a$, find the value of a .

Name	Value
Parabola	$\mathbf{x}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - (1 \ 0) \mathbf{x} = 0$
Line1	$\mathbf{x} = \begin{pmatrix} a \\ 0 \end{pmatrix} + \kappa_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
Line2	$\mathbf{x} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + \kappa_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Table: Parabola and Lines

Equating Area and Finding a

On comparing Parabola with

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (4.1)$$

The parameters of the Parabola are

$$\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u} = \frac{-1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, f = 0 \quad (4.2)$$

For the line $x - a = 0$, the parameters are

$$\mathbf{h}_1 = \begin{pmatrix} a \\ 0 \end{pmatrix}, \mathbf{m}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (4.3)$$

For the line $x - 4 = 0$, the parameters are

$$\mathbf{h}_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \mathbf{m}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (4.4)$$

The points of intersection for a conic and a line is given by

$$\kappa_i = \frac{1}{\mathbf{m}^T \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^T \mathbf{V} \mathbf{m})} \right) \quad (4.5)$$

$$g(\mathbf{x}) = \mathbf{x}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - (1 \ 0) \mathbf{x} \quad (4.6)$$

$$g(\mathbf{h}) = \mathbf{h}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h} - (1 \ 0) \mathbf{h} \quad (4.7)$$

$$\kappa_i = \left(0 \pm \sqrt{(1 \ 0) \mathbf{h} - \mathbf{h}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h}} \right) \quad (4.8)$$

$$\kappa_1 = \sqrt{a}, -\sqrt{a} \text{ and } \kappa_2 = 2, -2 \quad (4.9)$$

Therefore the points of intersection are

For line 1

$$\mathbf{P}_1 = \begin{pmatrix} a \\ \sqrt{a} \end{pmatrix}, \mathbf{P}_2 = \begin{pmatrix} a \\ -\sqrt{a} \end{pmatrix} \quad (4.10)$$

For line 2

$$\mathbf{P}_3 = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \mathbf{P}_4 = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \quad (4.11)$$

The area of the parabola in between the lines $x = a$ and $x = 4$ is given by

$$2 \left(\int_0^4 \sqrt{x} \, dx - \int_0^a \sqrt{x} \, dx \right) = \frac{4}{3} \left(8 - a^{\frac{3}{2}} \right) \quad (4.12)$$

The area between parabola and line $x = a$ is given by

$$2 \left(\int_0^a \sqrt{x} \, dx \right) = \frac{4}{3} a^{\frac{3}{2}} \quad (4.13)$$

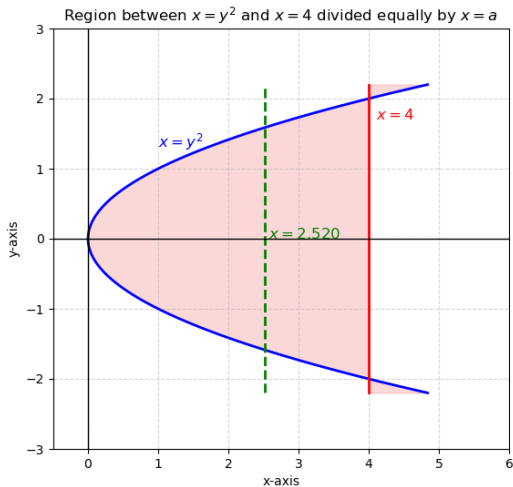
Given both the areas are equal

$$\frac{4}{3} \left(8 - a^{\frac{3}{2}} \right) = \frac{4}{3} a^{\frac{3}{2}} \quad (4.14)$$

$$4 = a^{\frac{3}{2}} \quad (4.15)$$

$$a = 4^{\frac{2}{3}} = 16^{\frac{1}{3}} \quad (4.16)$$

Plots



Figure

C Code

```
#include <stdio.h>
#include <math.h>

int main() {
    double total_area, half_area, a, lhs;

    // Step 1: Find total area between  $x = y^2$  and  $x = 4$ 
    // Limits are  $y = -2$  to  $y = 2$ 
    total_area = (4*2 - pow(2,3)/3) - (4*(-2) - pow(-2,3)/3);
    // Simplify the above: total_area = 32/3
    printf("Total-area-between- $x=y^2$ -and- $x=4$ :-%.6lf\n", total_area);

    // Step 2: Half of the total area
    half_area = total_area / 2.0;
    printf("Half-of-total-area:-%.6lf\n", half_area);

    // Step 3: For  $x = a$  dividing the region into equal parts:
```

```

// Area between  $x = y^2$  and  $x = a$  is  $(4/3) * a^{(3/2)}$ 
// We set  $(4/3)*a^{(3/2)} = \text{half\_area}$ 
//  $\Rightarrow a^{(3/2)} = (3/4)*\text{half\_area}$ 
double rhs = (3.0/4.0) * half_area;

// Now find  $a = (\text{rhs})^{(2/3)}$ 
a = pow(rhs, 2.0/3.0);

// Step 4: Display results
printf("Equation used:  $(4/3)*a^{(3/2)} = %.6lf\\n$ ", half_area);
printf(" $\Rightarrow a^{(3/2)} = %.6lf\\n$ ", rhs);
printf(" $\Rightarrow a = (%.6lf)^{(2/3)}\\n$ ", rhs);
printf("Value of a = %.6lf\\n", a);

return 0;
}

```

```
import ctypes

# Load shared object
lib = ctypes.CDLL("./area_equal_division.so")

# Alternatively, define a function in C to return 'a' value only.

# Let's assume we modify the C file to have find_a() returning value.
lib.find_a = lib.find_a if hasattr(lib, 'find_a') else None

if lib.find_a:
    lib.find_a.restype = ctypes.c_double
    print("The value of a =", lib.find_a())
else:
    print("Run 'area_equal_division' directly as a C-executable.")
```

Python Code for Plotting

```
import numpy as np
import matplotlib.pyplot as plt

# Function definitions
def parabola_x(y):
    return y**2
# Compute  $a = 4^{(2/3)}$ 
a = 4**(2/3)

# y values for plotting
y = np.linspace(-2.2, 2.2, 400)
# x values for curves
x_parabola = parabola_x(y)
x_vertical = 4 * np.ones_like(y)
x_dividing = a * np.ones_like(y)
```

```

# Plot settings
plt.figure(figsize=(7,6))
plt.plot(x_parabola, y, 'b', linewidth=2, label='$x = y^2$')
plt.plot(x_vertical, y, 'r', linewidth=2, label='$x = 4$')
plt.plot(x_dividing, y, 'g--', linewidth=2, label=fr'$x = a = \{a:.3f\}$')

# Fill area between  $x = y^2$  and  $x = 4$ 
plt.fill_betweenx(y, x_parabola, x_vertical, color='lightcoral', alpha=0.3)

# Labels near the lines (not overlapping)
plt.text(1.0, 1.3, '$x = y^2$', color='b', fontsize=12)
plt.text(4.1, 1.7, '$x = 4$', color='r', fontsize=12)
plt.text(a + 0.05, 0.0, fr'$x = \{a:.3f\}$', color='g', fontsize=12)

# Axes and limits
plt.axhline(0, color='k', linewidth=1)
plt.axvline(0, color='k', linewidth=1)
plt.xlim(-0.5, 6.0)
plt.ylim(-3.0, 3.0)

```

```
# Title and labels
plt.title("Region between  $x = y^2$  and  $x = 4$  divided equally by  $x = a$ ")
plt.xlabel("x-axis")
plt.ylabel("y-axis")

# Aspect ratio equal
plt.gca().set_aspect('equal', adjustable='box')

plt.grid(True, linestyle='--', alpha=0.5)
plt.savefig("../figs/fig15.png")
plt.show()
```